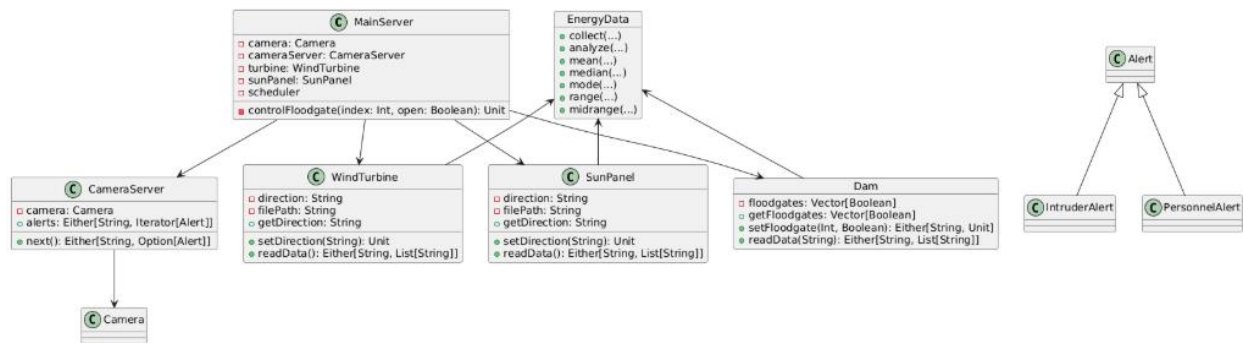


Part A:

Project Repository: <https://github.com/e-j-solutions/FunctionalProgrammingProject/tree/main>

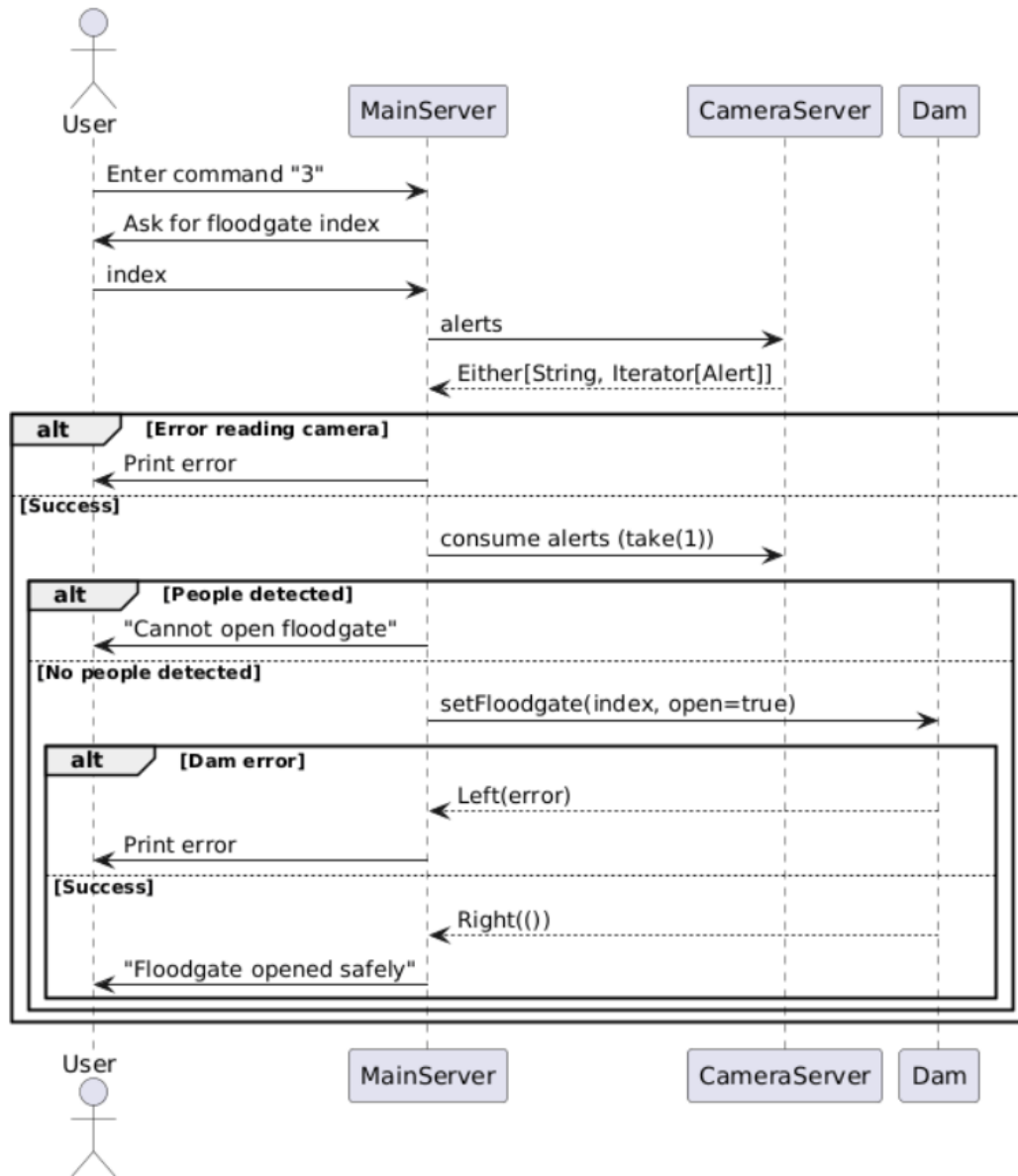
Video Explanation: [5_3_2026, 11_18_00 PM - Screen - Video Project 5.webm](#)

Class Diagram:

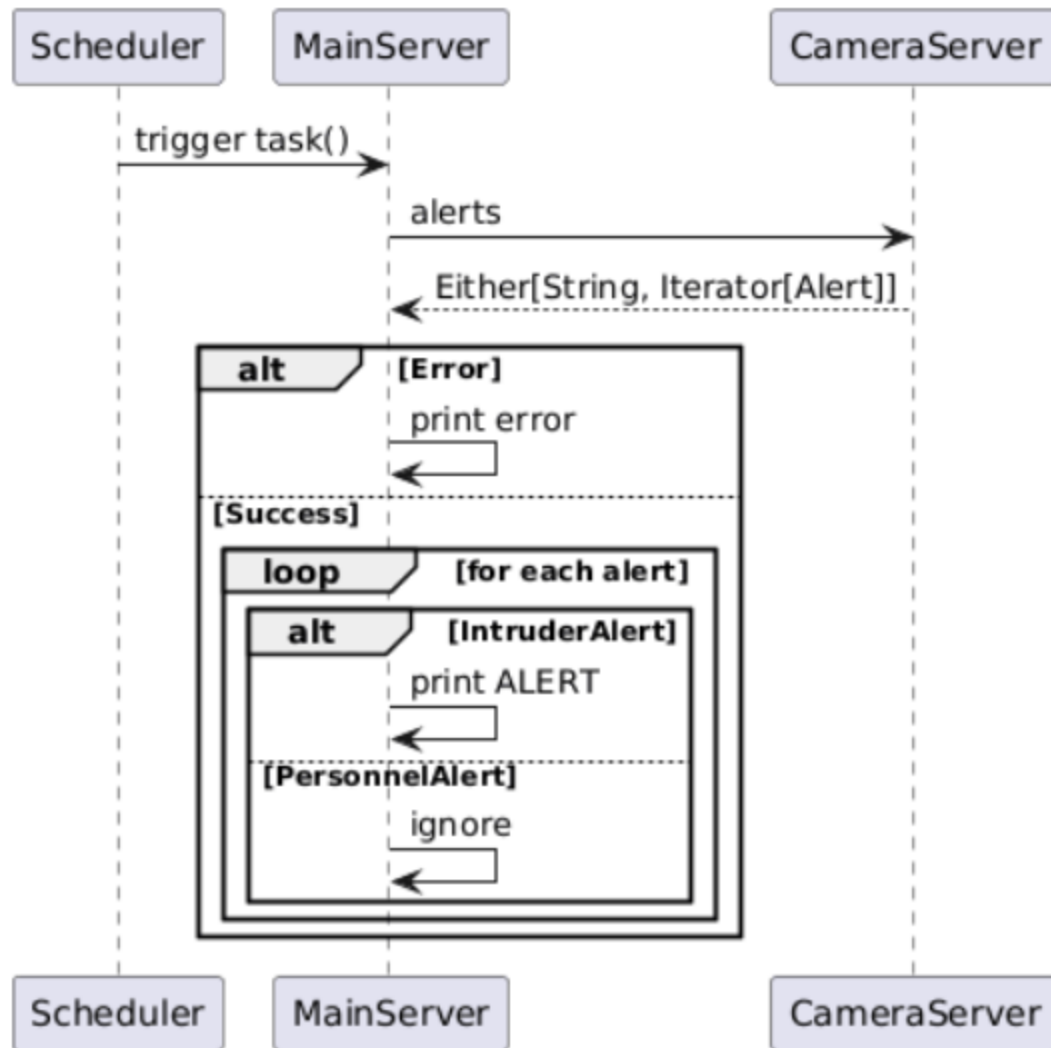


Sequence diagrams:

You want to open the floodgate (option 3):



Passive camera check:



Part B:

I chose to research Strictness and Laziness in Functional Programming. These terms refer to the way the programming language evaluates variables during compilations.

Strict(eager) languages evaluate variable the moment it is defined, doesn't matter whether it is going to be used in code or not. Lazy languages evaluate variables only when they are required to run the code. Technically, every language can be either strict (Scala) or lazy (Haskell), though usually there are ways to override it. For example, lazy val in Scala.

Example code can be found at <https://github.com/e-j-solutions/FunctionalProgrammingProject/blob/main/Example.scala>

References:

<https://www.tweag.io/blog/2022-05-12-strict-vs-lazy/>

https://en.wikipedia.org/wiki/Lazy_evaluation

https://en.wikipedia.org/wiki/Evaluation_strategy#Eager_evaluation